GRADE 10 / Physical Sciences / English

Motion in one dimension

01 Understand

Learning goals: Distinguish distance from displacement • Calculate average velocity • Interpret a straight position–time graph

Distance is total path length, a scalar. Displacement is final position minus initial position, a vector with a chosen positive direction.

Average velocity equals displacement divided by elapsed time. Average speed uses total distance instead.

On a position–time graph, gradient represents velocity. A horizontal line means the position is constant, not necessarily zero.

A walker returns towards the start; east is positive

Time (s)	Position (m east)	Interval velocity
0	0	—
2	4	+2 m·s^{-1}
4	0	−2 m·s^{-1}

02 Follow the reasoning

The walker moves 4 m east in 2 s, then 4 m west in 2 s. Find average speed and velocity.

1. Distance = 4 + 4 = 8 m; displacement = 0 − 0 = 0 m.

2. Elapsed time = 4 s, so average speed = 8 ÷ 4 = 2 m·s^{-1}.

3. Average velocity = 0 ÷ 4 = 0 m·s^{-1}. Motion occurred even though net displacement is zero.

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Practise with purpose

03 Try before checking

Use paper or your notebook. Show your reasoning. For written responses, use evidence and explain your choices. Suggested time: 15–20 minutes.

1. A cyclist travels 30 m east in 5 s. Find average velocity.

2. Velocity changes from 2 to 8 $m \cdot s^{-1}$ east in 3 s. Find acceleration.

3. Explain the gradient of a downward-sloping position–time line.

04 Check and correct

1. 6 $m \cdot s^{-1}$ east.

2. $a = (8 - 2)/3 = 2\ m \cdot s^{-2}$ east.

3. Velocity is negative relative to the chosen positive direction; it does not automatically mean slowing down.

Common mistake to avoid

Use $m \cdot s^{-2}$ for acceleration. A negative velocity describes direction, not automatically a decrease in speed.

Reflect and revisit

Identify one correction in your own words. Attempt a similar question tomorrow without looking at the guide. If you are still unsure, take your attempt to a teacher or tutor.

Coverage and sources

CAPS topic focus: Mechanics: instantaneous speed and velocity; acceleration

DBE CAPS, Grades 10–12: Physical Sciences. Section 3 content and skills. This is a focused supplement, not a complete syllabus or prescribed literature replacement.

Open the official DBE curriculum reference

Siyavula Grade 10 Physical Sciences Learner Book, chapter 21: speed and velocity (indexed PDF page 412; printed page 396). Available in the signed-in resource search. Original publisher attribution retained.

Content and visuals: original SmartBridge FET. AI-assisted authoring; subject-teacher and language-specialist review pending. Report an issue with the pack ID phsc-10.